GOVERNMENT ENGINEERING COLLEGE, SIWAN
B. Tech 2nd Semester (CSE & CSE-IOT), Mid-Sem Exam, 2025
Subject: Math-II

Time: 1.30 hours M.M: 20 Marks

Instructions:

(i) The marks are indicated in the right hand margin.
(ii) There are **SIX** questions in this paper.
(iii) Attempt **FOUR** questions in all.
(iv) Question No. **1** is compulsory.

Q.1: Choose the correct answer of the following: **(1×5=5)**

a) The PI of $(D^2+1)y = x$ is
(i) x
(ii) x^2
(iii) 1
(iv) None of these

b) The degree of the DE $\left(\frac{d^3y}{dx^3}+1\right)^{\frac{3}{2}} = x\frac{dy}{dx}+y$ is
(i) 1
(ii) 2
(iii) 3
(iv) 4

c) The PI of the $(D^2+4)y = e^{2x}$ is
(i) $\frac{e^{2x}}{4}$
(ii) $\frac{e^{2x}}{8}$
(iii) e^{2x}
(iv) $\frac{e^{2x}}{6}$

d) The infinite series $\sum_{n=1}^{\infty}\frac{1}{n^2}$ is
(i) Convergent
(ii) Divergent
(iii) oscillatory
(iv) None of there

e) Solve $\frac{dy}{dx} = xy$.

(f) Check the convergence of Sequence $\frac{1}{n}$

Q.2: Solve $\frac{d^2y}{dx^2} - 7\frac{dy}{dx} + 10y = e^{2x}\sin(x)$. **(5)**

Q.3: Solve Cauchy Euler linear differential equation $x^2\frac{d^2y}{dx^2} - x\frac{dy}{dx} + y = log(x)$. **(5)**

Q.4: Solve the following differential equation by variation of parameter method

$$\frac{d^2y}{dx^2} + y = cosec(x).$$ **(5)**

Q.5: Test the convergence of the infinite series **(5)**

$$\frac{x}{1.2} + \frac{x^2}{3.4} + \frac{x^3}{5.6} + \frac{x^4}{7.8} + ------$$

Q.6: Test the convergence of the infinite series **(5)**

$$\left(\frac{1}{3}\right)^2 + \left(\frac{1.2}{3.5}\right)^2 + \left(\frac{1.2.3}{3.5.7}\right)^2 + -------$$